Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.5

1. Determine if the system has a nontrivial solution. Try to use as few row operations as possible.

$$\begin{aligned} 4x_1 - 2x_2 + 10x_3 &= 0 \\ -4x_1 - 6x_2 - 4x_3 &= 0 \\ 8x_1 + 4x_2 + 14x_3 &= 0 \end{aligned}$$

Choose the correct answer below.

- **A.** The system has a nontrivial solution.
- **B.** It is impossible to determine.
- **C.** The system has only a trivial solution.

2. Determine if the system has a nontrivial solution. Try to use as few row operations as possible.

$$\begin{aligned} x_1 - 4x_2 + 6x_3 &= 0 \\ -2x_1 + 3x_2 - 2x_3 &= 0 \\ x_1 + x_2 + 7x_3 &= 0 \end{aligned}$$

Choose the correct answer below.

- **A.** The system has only a trivial solution.
- **B.** The system has a nontrivial solution.
- **C.** It is impossible to determine.

3. Write the solution set of the given homogeneous system in parametric vector form.

$$\begin{aligned} 4x_1 + 4x_2 + 8x_3 &= 0 \\ -12x_1 - 12x_2 - 24x_3 &= 0 \\ -5x_2 - 5x_3 &= 0 \end{aligned}$$

where the solution set is $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$\mathbf{x} = x_3 \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$

(Type an integer or simplified fraction for each matrix element.)

4. Describe all solutions of A**x** = **0** in parametric vector form, where A is row equivalent to the given matrix.

$$\begin{bmatrix} 1 & -4 & -9 & 5 \\ 0 & 1 & 3 & -2 \end{bmatrix}$$

$\mathbf{x} = x_3 \begin{bmatrix} -3 \\ -3 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix}$

(Type an integer or fraction for each matrix element.)

5. Describe all solutions of A**x** = **0** in parametric vector form, where A is row equivalent to the given matrix.

$$\begin{bmatrix} 1 & 3 & 0 & -5 \\ 4 & 12 & 0 & -20 \end{bmatrix}$$

$$\mathbf{x} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

(Type an integer or fraction for each matrix element.)

6. Describe and compare the solution sets of $x_1 + 3x_2 - 5x_3 = 0$ and $x_1 + 3x_2 - 5x_3 = -1$.

Describe the solution set, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, of $x_1 + 3x_2 - 5x_3 = 0$ in parametric vector form. Select the correct choice below and fill in the answer boxes within your choice.
(Type an integer or fraction for each matrix element.)

- A. $\mathbf{x} =$
- B. $\mathbf{x} = x_3$
- C. $\mathbf{x} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$
- D. $\mathbf{x} =$ $+ x_2$

Describe the solution set, $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, of $x_1 + 3x_2 - 5x_3 = -1$ in parametric vector form. Select the correct choice below and fill in the answer boxes within your choice.
(Type an integer or fraction for each matrix element.)

- A. $\mathbf{x} =$
- B. $\mathbf{x} =$ $+ x_3$
- C. $\mathbf{x} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 5 \\ 0 \\ 1 \end{bmatrix}$
- D. $\mathbf{x} = x_2$ $+ x_3$

Which option best compares the two equations?

- A. The solution set of the second equation is a plane parallel to the line that is the solution set of the first equation.
- B. The solution set of the second equation is a plane parallel to the plane that is the solution set of the first equation.
- C. The solution set of the second equation is a plane perpendicular to the line that is the solution set of the first equation.
- D. The solution set of the second equation is a line parallel to the line that is the solution set of the first equation.

7. Determine whether the statement below is true or false. Justify the answer.

A homogeneous equation is always consistent.

Choose the correct answer below.

- **A.** The statement is true. A homogenous equation can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. Such a system A**x** = **0** always has at least one nontrivial solution. Thus a homogenous equation is always consistent.
- **B.** The statement is true. A homogenous equation can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. Such a system A**x** = **0** always has at least one solution, namely, **x** = **0**. Thus a homogenous equation is always consistent.
- **C.** The statement is false. A homogenous equation can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. Such a system A**x** = **0** always has at least one nontrivial solution. Thus a homogenous equation is always inconsistent.
- **D.** The statement is false. A homogenous equation can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. Such a system A**x** = **0** always has at least one solution, namely, **x** = **0**. Thus a homogenous equation is always inconsistent.

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8. Determine whether the statement below is true or false. Justify the answer.

If **x** is a nontrivial solution of A**x** = **0**, then every entry in **x** is nonzero.

Choose the correct answer below.

- **A.** The statement is true. A nontrivial solution of A**x** = **0** is a nonzero vector **x** that satisfies A**x** = **0**. Thus, a nontrivial solution **x** can have some zero entries so long as not all of its entries are zero.
- **B.** The statement is false. A nontrivial solution of A**x** = **0** is a nonzero vector **x** that satisfies A**x** = **0**. Thus, a nontrivial solution **x** can have some zero entries so long as not all of its entries are zero.
- **C.** The statement is true. A nontrivial solution of A**x** = **0** is a nonzero vector **x** that satisfies A**x** = **0**. Thus, a nontrivial solution **x** cannot have any zero entries.
- **D.** The statement is false. The only solution of A**x** = **0** is the zero vector. Thus, a nontrivial solution does not exist.

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9. Determine whether the statement below is true or false. Justify the answer.

The homogenous equation A**x** = **0** has the trivial solution if and only if the equation has at least one free variable.

Choose the correct answer below.

- **A.** The statement is true. The homogenous equation A**x** = **0** has the trivial solution if and only if the equation has at least one free variable which implies that the equation has a nontrivial solution.
- **B.** The statement is false. The homogeneous equation A**x** = **0** never has the trivial solution.
- **C.** The statement is false. The homogeneous equation A**x** = **0** always has the trivial solution.
- **D.** The statement is true. The homogeneous equation A**x** = **0** has the trivial solution if and only if the matrix A has a row of zeros which implies the equation has at least one free variable.

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10. Determine whether the statement below is true or false. Justify the answer.

The equation A**x** = **b** is homogeneous if the zero vector is a solution.

Choose the correct answer below.

- **A.** The statement is true. A system of linear equations is said to be homogeneous if it can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. If the zero vector is a solution, then **b** = A**x** = A**0** = **0**.
- **B.** The statement is false. A system of linear equations is said to be homogeneous if it can be written in the form A**x** = **b**, where A is an $m \times n$ matrix and **b** is a nonzero vector in $\mathbb{R}^m$. Thus, the zero vector is never a solution of a homogeneous system.
- **C.** The statement is true. A system of linear equations is said to be homogeneous if it can be written in the form A**x** = **b**, where A is
- **D.** The statement is false. A system of linear equations is said to be homogeneous if it can be written in the form A**x** = **0**, where A is an $m \times n$ matrix and **0** is the zero vector in $\mathbb{R}^m$. If the zero vector is a solution, then **b** = A**x** = A**0** = **0**, which is false.